

Probability and Statistics

BSCS 3rd Semester — Mid Syllabus Notes

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Lecture 01 — Introduction & Basic Definitions

Random Experiment

Random experiment

An action or process that can be repeated under the same conditions, but whose outcome **cannot be predicted with certainty**.

Example

Tossing a coin $\rightarrow$ H, T.

Sample Space

Sample space (S)

The set of all possible outcomes of an experiment.

Example

$$S = \{H, T\}$$

Event

Event

A **subset** of the sample space is called an event.

Tossing two coins

$$S = \{HH, HT, TH, TT\}$$

- $A = \{HH\} \rightarrow$ only heads appear
- $B = \{TT\} \rightarrow$ only tails appear
- $C = \{HT, TH\} \rightarrow$ both head and tail appear

Types of Events

There are two types of events:

1. Mutually exclusive events
2. Not mutually exclusive events

Mutually Exclusive Events

Definition

Two events are mutually exclusive if they **cannot occur at the same time**.

Example

- When a coin is tossed, head or tail appears, but not both.
- Rolling a die:
 - $S = \{1, 2, 3, 4, 5, 6\}$
 - $A = \{1, 3, 5\}$
 - $B = \{2, 4, 6\}$
 - $A \cap B = \phi \rightarrow$ these are mutually exclusive (M.E.)

Not Mutually Exclusive Events

Definition

Two events are *not* mutually exclusive if they **can occur at the same time**.

Example

- $S = \{1, 2, 3, 4, 5, 6\}$
- $A = \{1, 3, 5, 6\}$
- $B = \{2, 4, 6\}$
- $A \cap B = \{6\}$

Lecture 02 & 03 — Probability

Probability

The measure of the chance or likelihood that a particular event occurs.

$$P(E) = \frac{n(A)}{n(S)}$$

- $n(A) \rightarrow$ number of outcomes in the **event**
- $n(S) \rightarrow$ number of outcomes in the **sample space**

Rolling a die — probability of an even number

- $S = \{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$
- $A = \{2, 4, 6\}$, so $n(A) = 3$

$$P(E) = \frac{3}{6} = \frac{1}{2} = 0.5$$

MCQ

Probability always lies between 0 and 1:

$$0 \leq P(A) \leq 1$$

Statistics — Basic Terms

Statistics

The branch of mathematics that deals with **collecting, analyzing, interpreting and presenting** data into useful information, or for making effective decisions.

Example

- Used in education to analyze student performance.
- Used in business.

Data

Definition

A collection of raw facts, numbers and observations.

Example

Marks of students.

Data Analysis

Definition

Studying the collected data.

Example

If you take the marks of students and find the average, you are doing data analysis.

Types of Statistics

1. Descriptive statistics
2. Inferential statistics

Descriptive Statistics

Used to **summarize and describe** the main features of data.

Example

Mean, median, mode.

Inferential Statistics

Used to **make conclusions about a large group** based on data from a sample.

Example

A man starting a chocolate business picks some people from a large population, has them taste his chocolate, and draws a conclusion from their responses.

Types of Data

1. Discrete data
2. Continuous data

Discrete Data

Definition

Countable data. It includes only whole numbers $\{0, 1, 2, 3, \dots\}$.

Example

Number of students in a class.

Continuous Data

Definition

Measurable data. It may be in decimal form.

Example

- Height of a student (e.g. 4.5 ft)
- Weight of a student (e.g. 60 kg)

Population

Definition

The entire group of individuals or items that you want to study.

Example

- All students of a school.
- All people living in Pakistan.

Worked examples — Dice and Coins

Q: Probability of two dice

Sample space (36 outcomes):

$$S = \left\{ \begin{array}{l} (1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), \\ (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), \\ (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), \\ (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), \\ (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), \\ (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6) \end{array} \right\}$$

Event A = both dice show the same number:

$$A = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

- $n(A) = 6$
- $n(S) = 36$

$$P(E) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6} \approx 0.17$$

Probability of two coins

- $S = \{HH, HT, TH, TT\}$, so $n(S) = 4$
- Event A = first coin shows head: $A = \{HH, HT\}$, so $n(A) = 2$

$$P(E) = \frac{n(A)}{n(S)} = \frac{2}{4} = \frac{1}{2} = 0.5$$

Lecture 04 — Conditional Probability

Conditional probability

The probability of an event occurring **when another event has already occurred**.

$P(A/B)$ = probability of A given that B has occurred.

Formula

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$P(A \cap B)$ = probability that **both** A and B occur.

Example — 3 fair coins tossed

Find the probability that:

- at least two tails appear
- the first coin shows head

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$, so $n(S) = 8$

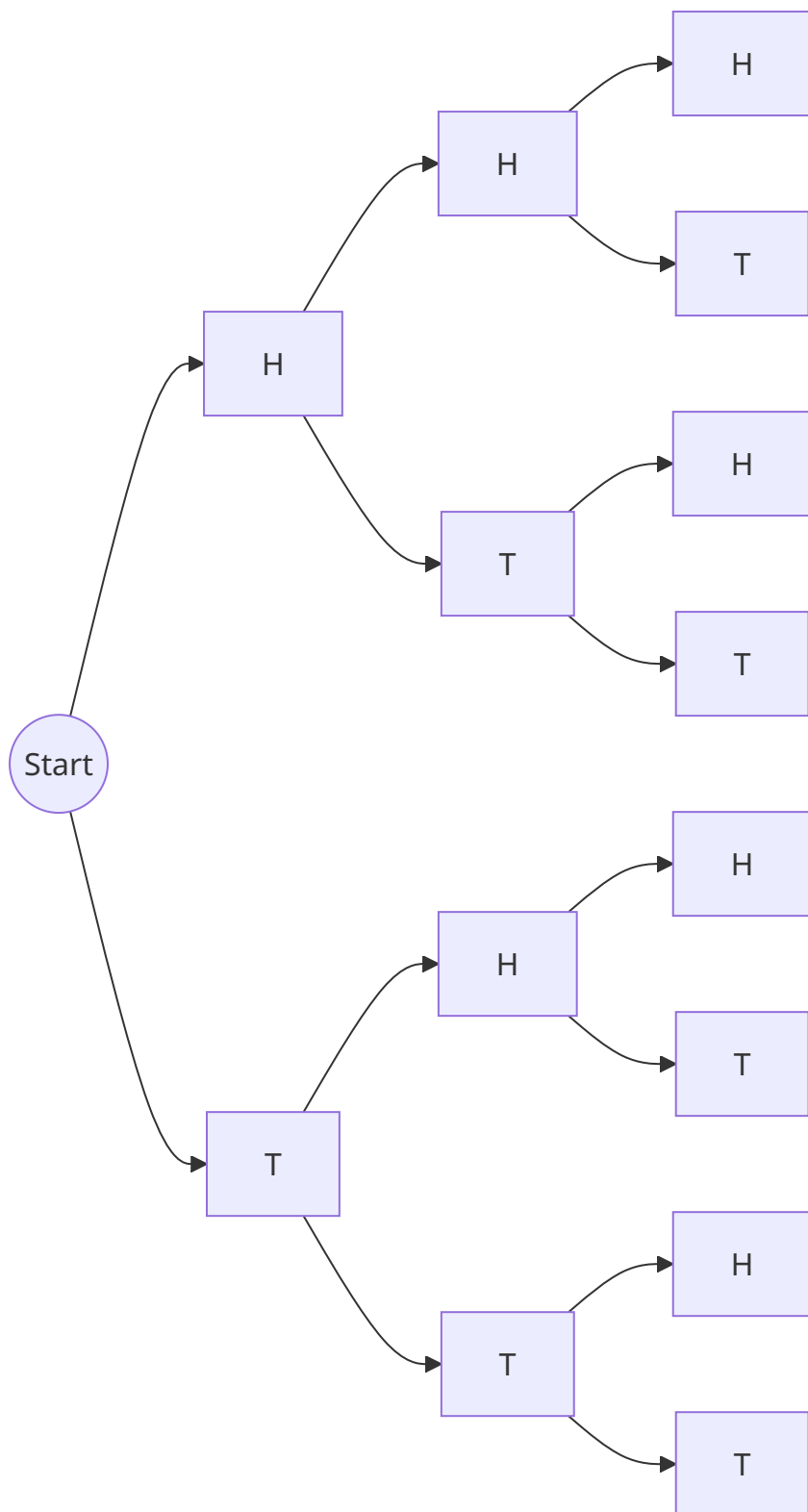
Event A — at least two tails: $A = \{HTT, THT, TTH, TTT\}$, $n(A) = 4$

$$P(A) = \frac{4}{8} = \frac{1}{2}$$

Event B — first coin shows head: $B = \{HHH, HHT, HTH, HTT\}$, $n(B) = 4$

$$P(B) = \frac{4}{8} = \frac{1}{2}$$

Tree method



Q: At least two tails appear, given that the first coin shows head

$A \cap B = \{HTT\}$, so $n(A \cap B) = 1$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{1}{8} = 0.125$$

Conditional probability:

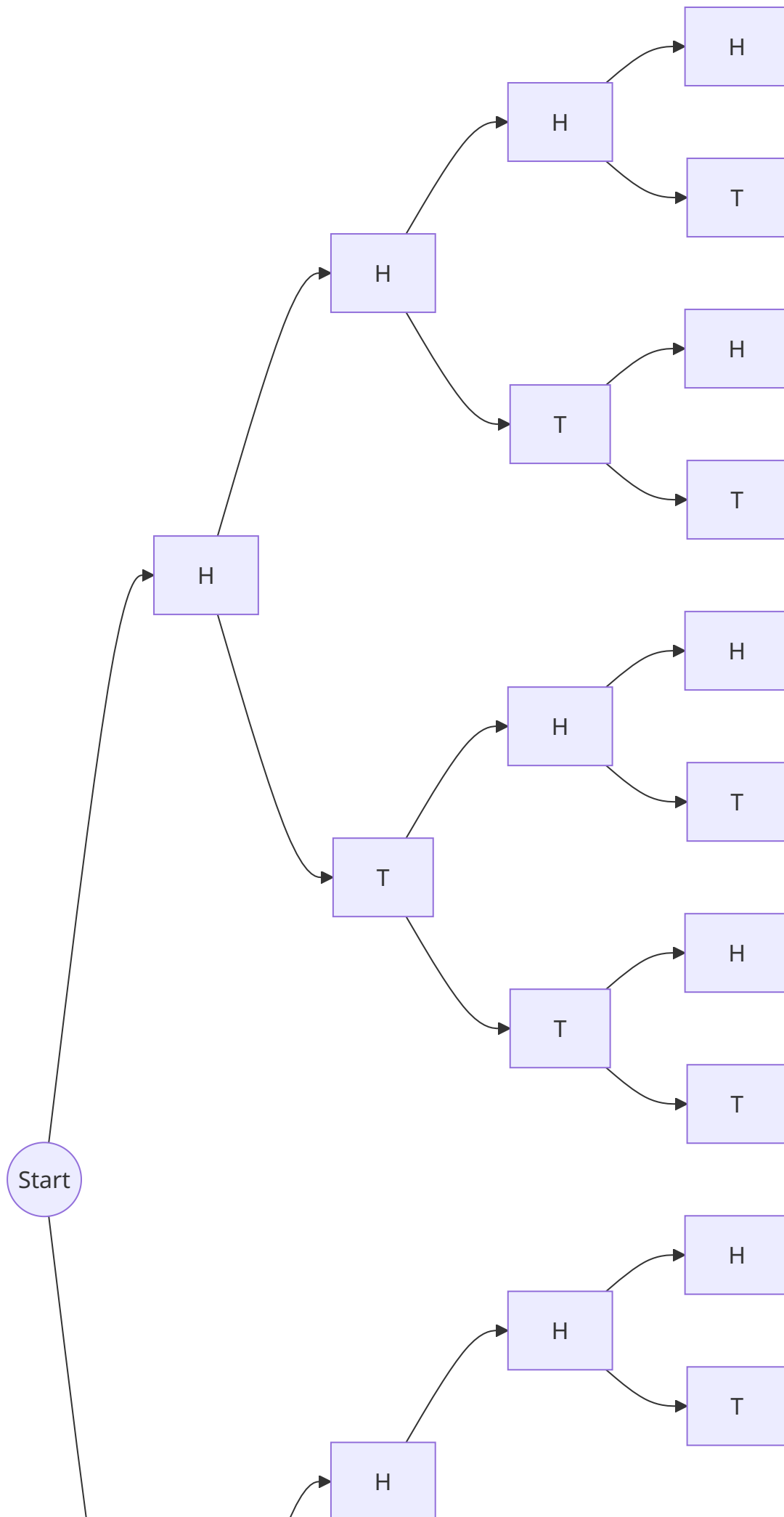
$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1/8}{1/2} = \frac{1}{4} = 0.25$$

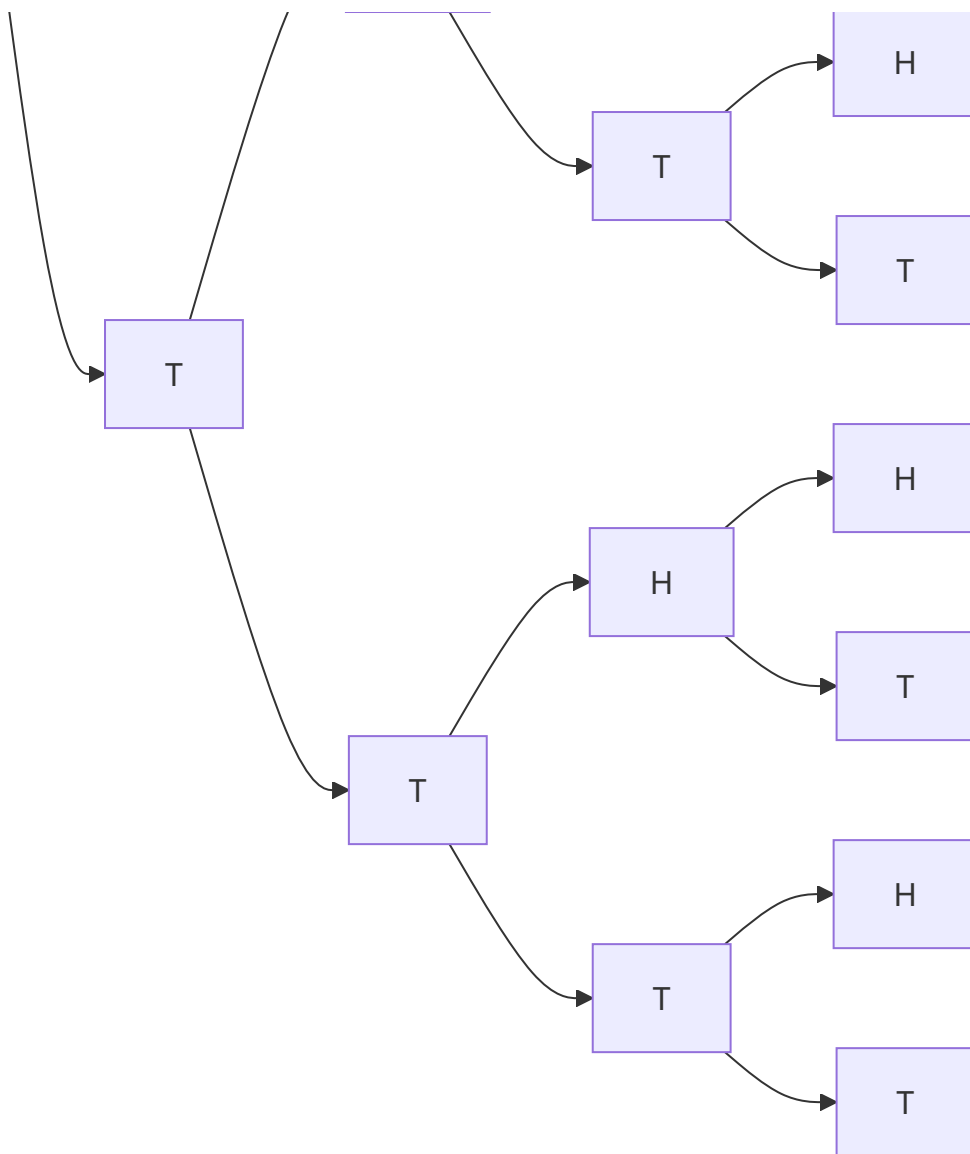
Example — 4 fair coins tossed

Find the probability that:

- at least three tails appear
- the first two coins show head

Tree method





$S = \{HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT, THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT\}$
 $n(S) = 16$

Event A — at least three tails: $A = \{HTTT, THTT, TTHT, TTTH, TTTT\}$, $n(A) = 5$

$$P(A) = \frac{5}{16} \approx 0.31$$

Event B — first two coins show head: $B = \{HHHH, HHHT, HHTH, HHTT\}$, $n(B) = 4$

$$P(B) = \frac{4}{16} = \frac{1}{4} = 0.25$$

Intersection: $A \cap B = \phi$, so $n(A \cap B) = 0$

$$P(A \cap B) = \frac{0}{16} = 0$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{0}{1/4} = 0$$

Lecture 05 & 06 — Probability Examples (Deck of Cards)

⋮ Problem

A card is drawn from an ordinary deck of 52 playing cards. Find the probability that:

1. the card is **red**
2. the card is a **diamond**
3. the card is a **10**

Deck facts

Colour	Suits	Cards
Red	13 Diamonds + 13 Hearts	26
Black	13 Spades + 13 Clubs	26

Each suit contains: 1 (Ace), 2, 3, 4, 5, 6, 7, 8, 9, 10, K, Q, J.

MCQs

- Red = 13 diamonds and 13 hearts → **total red = 26**
- Black = 13 spades and 13 clubs → **total black = 26**
- Face cards = **12** (K, Q, J in each of the 4 suits)
- Total cards = **52**

Solution

$$n(S) = 52$$

1. Card is red — let A be the event. $n(A) = 26$

$$P(A) = \frac{26}{52} = \frac{1}{2}$$

2. Card is a diamond — let B be the event. $n(B) = 13$

$$P(B) = \frac{13}{52} = \frac{1}{4} = 0.25$$

3. Card is a 10 — let C be the event. $n(C) = 4$

$$P(C) = \frac{4}{52} = \frac{1}{13} \approx 0.077$$

Multiplication Law

Important question

Multiplication Law / Rule / Product Law

Statement

If A and B are two events defined in sample space S , then

- $P(A \cap B) = P(A) \cdot P(B|A)$

- $P(A \cap B) = P(B) \cdot P(A/B)$

Proof

By the definition of conditional probability:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

$$\Rightarrow P(A \cap B) = P(B) \cdot P(A/B)$$

Now,

$$P(B/A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) \neq 0$$

$$\Rightarrow P(A \cap B) = P(A) \cdot P(B/A) \quad \blacksquare$$

Lecture 07 — Random Variable

Random variable / Chance variable

Definition

A variable whose value is determined by the outcome of a random experiment. It is also called a **stochastic variable**.

Two coins tossed

$$S = \{HH, HT, TH, TT\}$$

Assign a numerical value = number of heads:

Outcome	HH	HT	TH	TT
Value	2	1	1	0

So the variable takes values $i = 0, 1, 2$.

Counting Sample Points

Counting sample points is divided into three rules:

1. Rule of Multiplication
2. Rule of Permutation
3. Rule of Combination

Rule of Multiplication

Definition

If two experiments with m and n outcomes are performed, the total number of outcomes is

$$m \times n = mn$$

☰ Coin and die

- Coin: $m = \{H, T\}$, so $m = 2$
- Die: $n = \{1, 2, 3, 4, 5, 6\}$, so $n = 6$

By the multiplication rule: $m \times n = 2 \times 6 = 12$ outcomes.

Rule of Permutation

Definition

Used when the **order of selection matters**.

$${}_nP_r = \frac{n!}{(n-r)!}$$

- $n \rightarrow$ total observations
- $r \rightarrow$ number selected

☰ Club with 4 members

Choose a President, Secretary and Treasurer: $n = 4, r = 3$

$${}_4P_3 = \frac{4!}{(4-3)!} = \frac{4!}{1!} = \frac{4 \cdot 3 \cdot 2 \cdot 1}{1} = 24$$

Rule of Combination

Definition

Used when the **order of selection does not matter**.

$${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

☰ Committee

Choose a committee of $r = 3$ from $n = 4$:

$${}_4C_3 = \frac{4!}{3!(4-3)!} = \frac{4!}{3! \cdot 1!} = \frac{24}{6} = 4$$

Additive Law

Statement

If A and B are two events (not mutually exclusive) defined in S , then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof

The event $A \cup B$ can be written as:

$$\begin{aligned} A \cup B &= A \cup (\bar{A} \cap B) \\ P(A \cup B) &= P(A) + P(\bar{A} \cap B) \end{aligned} \quad (1)$$

Again, event B decomposes as:

$$\begin{aligned} B &= (A \cap B) \cup (\bar{A} \cap B) \\ P(B) &= P(A \cap B) + P(\bar{A} \cap B) \end{aligned} \quad (2)$$

Subtract (2) from (1):

$$\begin{aligned} P(A \cup B) - P(B) &= P(A) + P(\bar{A} \cap B) - P(A \cap B) - P(\bar{A} \cap B) \\ \Rightarrow P(A \cup B) &= P(A) + P(B) - P(A \cap B) \quad \blacksquare \end{aligned}$$

Lecture 09 — Discrete and Continuous Random Variables

Discrete Random Variable

Takes whole-number (countable) values.

Quantity	Formula
Mean	$E(X) = \sum x p$
$E(X^2)$	$E(X^2) = \sum x^2 p$
Variance	$\text{Var}(X) = E(X^2) - [E(X)]^2$
Std. deviation	S.D = $\sqrt{\text{Var}(X)}$

Example 1

A discrete random variable has the probabilities below:

X	-2	-1	0	1	2	3
$P(X)$	0.2	k	0.1	$2k$	0.1	$2k$

Find k (since $\sum p = 1$):

$$\begin{aligned} 0.2 + k + 0.1 + 2k + 0.1 + 2k &= 1 \\ 0.4 + 5k &= 1 \Rightarrow 5k = 0.6 \Rightarrow k = \frac{0.6}{5} = 0.12 \end{aligned}$$

Mean:

$$E(X) = \sum xp = (-2)(0.2) + (-1)(0.12) + (0)(0.1) + (1)(0.24) + (2)(0.1) + (3)(0.24)$$

$$\boxed{\text{Mean} = 0.64}$$

Variance:

$$E(X^2) = \sum x^2p = 3.72$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 3.72 - (0.64)^2$$

$$\boxed{\text{Variance} = 3.3104}$$

Standard deviation:

$$\text{S.D} = \sqrt{3.3104} \approx \boxed{1.819}$$

Example 2

X	0	1	2
$P(X)$	1/4	1/2	1/4

Mean:

$$E(X) = (0)\frac{1}{4} + (1)\frac{1}{2} + (2)\frac{1}{4} = \boxed{1}$$

Variance:

$$E(X^2) = \sum x^2p = 0 + \frac{1}{2} + 1 = 1.5$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 1.5 - 1 = 0.5 \Rightarrow \boxed{\text{Var} = \frac{1}{2}}$$

Standard deviation:

$$\text{S.D} = \sqrt{\text{Var}(X)} = \sqrt{\frac{1}{2}} = \boxed{\frac{1}{\sqrt{2}}}$$

Continuous Random Variable

Quantity	Formula
Mean	$E(X) = \int x f(x) dx$
$E(X^2)$	$E(X^2) = \int x^2 f(x) dx$
Variance	$\text{Var}(X) = E(X^2) - [E(X)]^2$

Example

$$f(x) = k(x - x^2), \quad 0 \leq x \leq 1$$

Find: **k, Mean, Variance.**

Find k (total area must equal 1):

$$k \int_0^1 (x - x^2) dx = 1$$

$$k \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = 1$$

$$k \left(\frac{1}{2} - \frac{1}{3} \right) = 1 \Rightarrow k \left(\frac{3-2}{6} \right) = 1 \Rightarrow k \cdot \frac{1}{6} = 1$$

$$\boxed{k = 6}$$

Mean:

$$E(X) = \int_0^1 x f(x) dx = \int_0^1 x \cdot 6(x - x^2) dx = 6 \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = 6 \cdot \frac{1}{12} = \frac{1}{2}$$

Variance:

$$E(X^2) = \int_0^1 x^2 \cdot 6(x - x^2) dx = 6 \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 = \frac{3}{10}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = \frac{3}{10} - \left(\frac{1}{2} \right)^2 = \frac{3}{10} - \frac{1}{4}$$

$$\boxed{\text{Var} = \frac{1}{20}}$$

Lecture 10 — Bayes' Theorem

Statement

If the events $A_1, A_2, A_3, \dots, A_k$ form a **partition** of a sample space S — that is, the events A_i are mutually exclusive and their union is S , such that B can occur only if one of the A_i occurs — then for any i :

$$P(A_i/B) = \frac{P(A_i) P(B/A_i)}{\sum_{i=1}^k P(A_i) P(B/A_i)}, \quad i = 1, 2, \dots, k$$

Proof

By the definition of conditional probability, for events A_i and B :

$$P(A_i \cap B) = P(A_i) \cdot P(B/A_i) \quad (\text{i})$$

$$P(A_i \cap B) = P(B) \cdot P(A_i/B) \quad (\text{ii})$$

Comparing (i) and (ii):

$$P(A_i) \cdot P(B/A_i) = P(B) \cdot P(A_i/B)$$

$$P(A_i/B) = \frac{P(A_i) \cdot P(B/A_i)}{P(B)} \quad (\text{A})$$

Now write event B :

$$B = S \cap B = (A_1 \cup A_2 \cup A_3 \cup \dots \cup A_k) \cap B$$

$$B = (A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_k \cap B)$$

Since the $A_i \cap B$ are mutually exclusive:

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + \dots + P(A_k \cap B)$$

$$P(B) = P(A_1) P(B/A_1) + \dots + P(A_k) P(B/A_k) = \sum_{i=1}^k P(A_i) P(B/A_i)$$

Substituting $P(B)$ into equation (A):

$$P(A_i/B) = \frac{P(A_i) \cdot P(B/A_i)}{\sum_{i=1}^k P(A_i) \cdot P(B/A_i)} \quad \blacksquare$$
